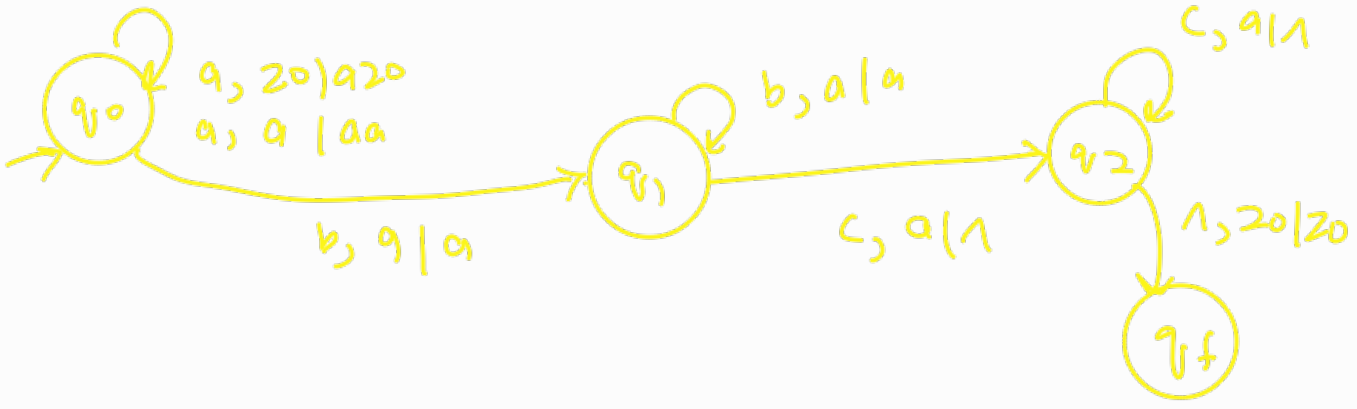
PDA example

$L_3 = \{ a^n b^m c^n \mid m, n >= 1 \}$

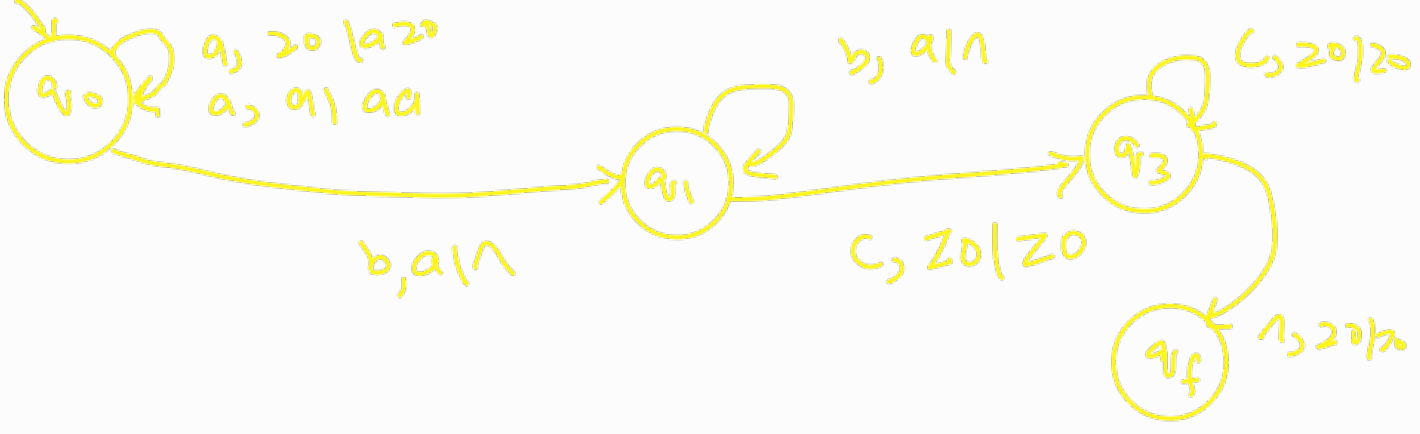
aaa bb ccc

push skip pop



$L_4 = \{ a^n b^n c^m \mid m, n >= 1 \}$

aaabbb cc



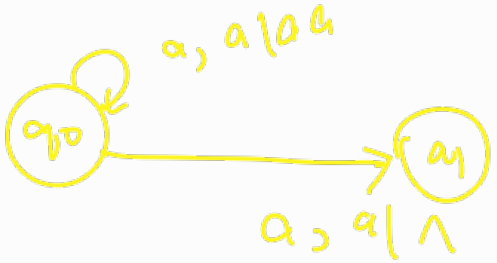
L_1 $L = \{ a^m b^n c^p d^q \mid m+n = p+q \}$

aa bb ccdd

abbb cccd

DPDA → there is only a single move

for a single alphabet we must move to only 1 stack



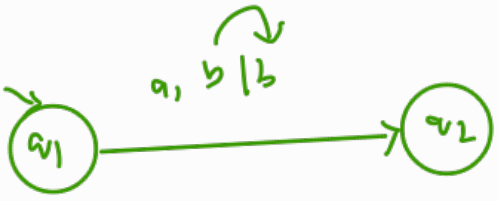
(X) non-deterministic

↘ here for 'a' we are moving to two states.

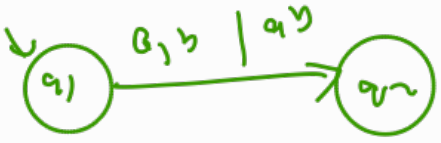
$L = a^n b^{2n} \quad n>0$

Stack transion function

δ(current state, input, TOS) = (next state, push/pop/skip)



$\delta(q_1, a, b) = (q_2, b)$



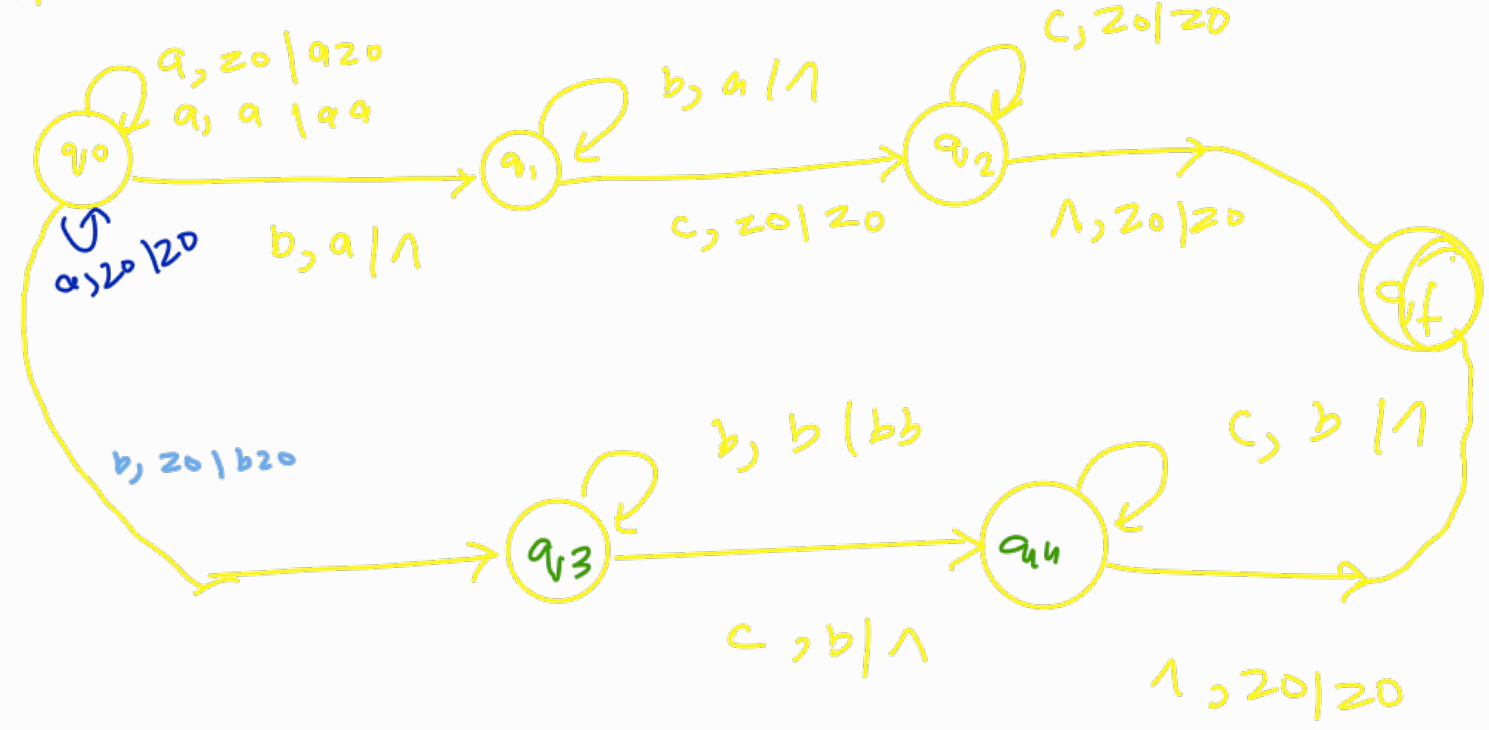
$\delta(q_1, a, b) = (q_2, ab)$

$L = \{ a^i b^j c^k \mid i=j \text{ or } j=k \}$

L_1 → aabbcccc

L_2 → aaabc

4



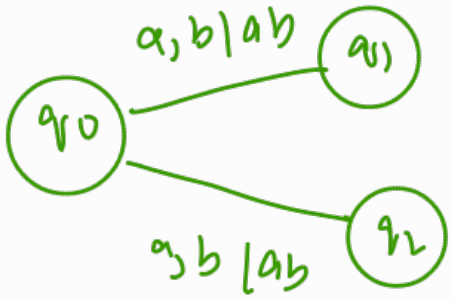
NPDA

Stack alphabets → {a, b, z0}

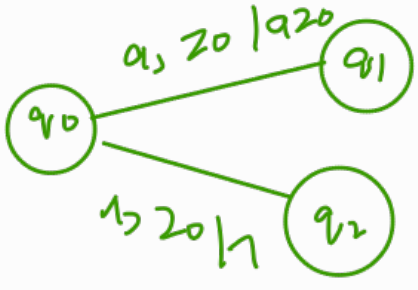
those we have pushed in the stack

NPDA

if two or edges labelled with same input



if a state has two edges with same stack symbol and one of input symbol is $\wedge$.

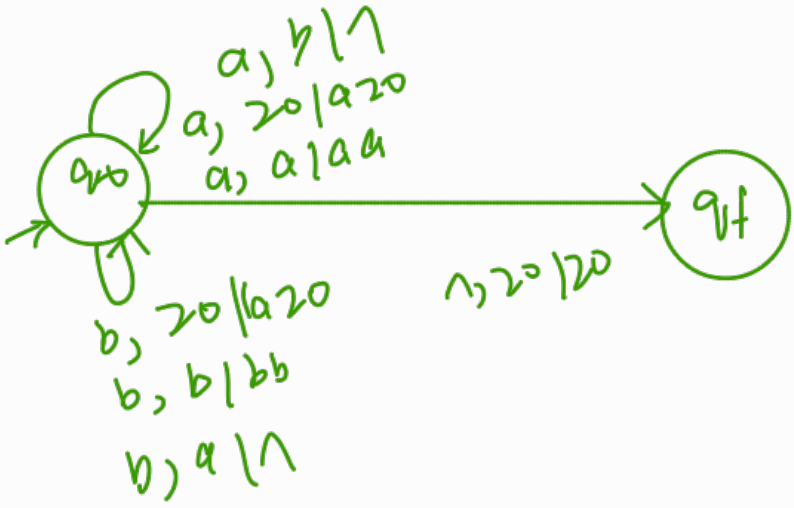


we can either consume input or not consume it.

$L = \{w,\ w \in (a,b)^*\ n_a = n_b\}$

for every a pushed pop b or

for every b pushed pop a.



CNF if all the productions are of form

$A \to BC$ or $A \to a$

where A, B, C are variables and a is a terminal

$A \to aB$ is not cnf.

and only Start variable can derive null: $A \to \epsilon$

Its derivation tree or parse tree is always a binary tree

no of steps required to derive a string of Length $|w|$ is $2|w| - 1$

$w = abcd$

$|w| = 4$

$2|4| - 1$

$\to (7)$ So in 7 steps we can drive a string in CNF

converting CFG → CNF

$A \to BC$

$A \to a$

$S \to ABa$

$A \to aab$

$B \to Ac$

wherever there is terminal, substitute using a var

$X \to a$

$Y \to b$

$Z \to C$

$S \to ABX$ or $S \to AB$

$A \to XXY$ → not in cnf

$B \to AZ$

$T \to AB$

$L \to XX$

$S \to TX$

$A \to LY$

$B \to AZ$

$X \to a$

$Y \to b$

$Z \to C$

Remove null → unit productio. → useless prod → CFG → CNF

$S \to bA \mid aB$

$A \to bAA \mid as \mid a$

$B \to aBB \mid bS \mid b$

$X \to b \qquad Y \to a$

$S \to XA \mid YB$

$A \to XAA \mid YS \mid a$

$B \to YBB \mid XS \mid b$

$P \to XA$

$R \to YB$

$S \to XA \mid YB$

$A \to PA \mid YS \mid a$

$B \to RB \mid XS \mid b$

(Removal of null production)

$S \to aAB$

$A \to aAA \mid \in$

$B \to bBB \mid \in$

null variable

→ variable that derive null directly or indirectly

$\{A, B\}$

if the production is in form

$A \to \Lambda$

↙ substitute null variable

$S \to aAB \mid aB \mid aA \mid a$

$A \to aAA \mid aA \mid a$

$B \to bBB \mid bB \mid b$

$S \to AB \mid B \mid A \mid \Lambda$

$A \to aAA \mid aA \mid a$

$B \to bBB \mid bB \mid b$

(null can only come with start)

$S \to \Lambda$

null variable

$\{A, B, S\}$

Q $S \to AbaC$

$A \to BC$

$B \to b \mid \epsilon$

$C \to D \mid \epsilon$

$D \to d$

nullable

$\{B, C, A\}$

checking if S is

$A \to \wedge\wedge$ ✓

$S \to \wedge ba \wedge$

(K)

$S \to Abac \mid bac \mid Aba \mid ba$

$A \to BC \mid B \mid C$

$B \to b$

$C \to D$

$D \to d$



(Removal of unit production)

$S \to Aa \mid B$
$B \to A \mid bb$
$A \to a \mid bc \mid B$

unitprod $S \to B$
unitprod $B \to A$
unitprod $A \to B$

$A \to B$
one variable produces exactly one variable
↓
unit production

Removing unit prod

$S \to Aa \mid bb \mid a \mid bc$
$B \to bb \mid a \mid bc$
$A \to a \mid bc \mid bb$

Don't write B again if it generates itself.

$S \to B \to A \to a \mid bc \mid B$
$\to bb$

$B \to A \to a \mid bc \mid B$

$A \to B \to A \mid bb$

$S \to AB$

$A \to a$

$B \to b \mid C$

$C \to D$

$D \to E$

$E \to a$

Unit

$B \to C$

$C \to D$

$D \to E$

$S \to AB$

$A \to a$

$B \to b \mid a$

$C \to a$

$D \to a$

$E \to a$

removed the production and replaced with strings

$B \to C \to D$

$\downarrow$

E

$\to a$

$C \to D \to E \to a$

$D \to E \to a$

useless production → they are not reachable

here C and D and E are not reachable. from staring state

(Removal of useless productions)

$S \to AB \mid a$ (X)
$A \to BC \mid b$
$B \to aB \mid C$ (X)
$C \to aC \mid B$

} X useless

useful Symbol → terminals and variables that derive them.

→ $\{a, b, S, A\}$

unit production: A → B means RHS and LHS must contain variable

$A \to B$

$S \to a$
$A \to b$ → (X)

as we cannot reach A from S

useless

$S \to a$ → trivial

useful Symbol

$\{a, b, A, B, S\}$

Take this check from this combination is where at DHS

$S \to AB \mid AC$ (X)
$A \to aAb \mid bAa \mid a$
$B \to bbA \mid aaB \mid AB$
$C \to abCA \mid aDb$ (X)
$D \to bD \mid aC$ (X)

$S \to AB$
$A \to aAb \mid bAa \mid a$
$B \to bbA \mid aaB \mid AB$

$S \to TU \mid V$

$T \to aTb \mid \wedge$

$U \to cU \mid \wedge$

$V \to aVc \mid W$

$W \to bW \mid \wedge$

$\{T, U, W, V\}$ → now

Substitute null production where these variables occur

$S \to TU \mid V \mid U \mid T$

$T \to aTb \mid ab$

$U \to cU \mid c$

$V \to aVc \mid W \mid ac$

$W \to bW \mid b$

Removing unit prod

$S \to U$

$S \to T$

$S \to V$

$V \to W$

Removing unit prod

$S \to U \to cU \mid c$

$S \to T \to aTb \mid ab$

$S \to V \to aVc \mid W \mid ac$

$V \to W \to bW \mid b$

$S \to TU \mid V \mid U \mid T$

$T \to aTb \mid ab$

$U \to cU \mid c$

$V \to aVc \mid W \mid ac$

$W \to bW \mid b$

Removal of useless

$S \to TU \mid aVc \mid ac \mid bW \mid b \mid cU \mid c$

$T \to aTb \mid ab$ (aTb, ab)

$U \to cU \mid c$

$V \to aVc \mid bW \mid b \mid ac$

$W \to bW \mid b$

Remove non-generating symbol.

→

production that don't produce terminals directly

(Remove useless productions)

Remove all productions that don't generate alone terminals.

$S \to a \mid bXY$ (X)

$A \to Bad \mid bsx \mid a$ (X)

✓ $B \to aSB \mid bBX$

$X \to SBD \mid aBX \mid ad$

✓ $Y \to SBa \mid XYb$

$S \to a$

$A \to bSX \mid a$ } (X) ($S \to a$)

$X \to ad$

$S \to AB \mid a$ (X)

$A \to BC \mid b$

(X) $B \to aB \mid C$

(X) $C \to ac \mid B$

$S \to a$ ✓

$A \to b$ (X)

(Practice :)

$S \to 0A \mid 1B \mid C$

$A \to 0S \mid 00$

$B \to 1 \mid A$

$C \to 01$

$S \to C$ unit prod..

$B \to A$

Eliminating unit prod

$S \to C$ $\quad C \to 02$

$S \to 01$ ✓

$B \to A$ $\quad A \to 0S \mid 00$

$B \to 0S \mid 00$ ✓

final Ans

$S \to 0A \mid 1B \mid 01$

$A \to 0S \mid 00$

$B \to 1 \mid 0S \mid 00$

$C \to 01$

e.g

$S \to AB$

$A \to a$

$B \to C \mid b$

$C \to D$

$D \to E \mid bC$

$E \to d \mid Ab$

(Unit productions)

$B \to C$

$C \to D$

$D \to E$

starting from D.

$D \to d \mid Ab \mid bC$

$C \to d \mid Ab \mid bC$ ✓

$B \to d \mid Ab \mid bC \mid b$

NUL-production is that product that contains $A \to \wedge$

$S \to xyx$
$X \to 0X \mid \wedge$
$Y \to 1Y \mid \wedge$

$X \to \wedge$
$Y \to \wedge$

Inplace of each X Substitute $\wedge$.

$S \to xyx \mid yx \mid xy \mid y$
$X \to 0X \mid 0$
$Y \to 1 \mid Y \mid$

Now in place of every Y place $\wedge$.

$S \to xyx \mid yx \mid xy \mid y \mid xx \mid x \mid \wedge$ ← ??
$X \to 0X \mid 0 \mid$
$Y \to 1Y \mid 1$

(Removal of useless Symbols)

$S \to AB$
$A \to a$

(B is useless)

what Symbol is useless
→ When it cannot be reached from start Symbol
→ It cannot derive any terminals / alone terminals

e.g $S \to XY \mid 0$

$X \to 1$

Remove $S \to XY$ as Y is useless because it doesn't have any terminals

$\begin{bmatrix} S \to 0 \\ X \to 1 \end{bmatrix}$ ✓

e.g $S \to A11B \mid 11A$

$S \to 11B \mid 11$

$A \to 0$

$B \to BB$

$B \to BB$ (X)

useless

Since B is useless every production associated with it is also useless

$S \to 11A$

$S \to 11$

$A \to 0$ ✓

Simplification of CFG → By elimination of useless prod
null prod
unit prod

Remove null → unit production → useless prod → CFG → CNF

e.g

$S \to A|0C1$
$A \to B|0|10$
$C \to \Lambda|CD$

Removing useless prod

$A \to B$ (✓)
$C \to CD$ (✗)

B is useless because it doesn't have any terminals.

$S \to A|0C1$
$A \to 01|10$
$C \to \Lambda$

(Removing null productions)

$C \to \Lambda$

$S \to A|0C1|01$
$A \to 01|10$

(Removing unit production)

$S \rightarrow A$ (unit production) $S \rightarrow 0C1$ $S \rightarrow 01$

$S \rightarrow 01 \mid 10 \mid 0C1$

$A \rightarrow 01 \mid 10$ ✓

(CNF) → when a cfg is said to be in $A \rightarrow BC$ or $A \rightarrow a$ only.

$S \rightarrow bA \mid aB$

$A \rightarrow bAA \mid aS \mid a$

$B \rightarrow aBB \mid bS \mid b$

CNF

$A \rightarrow a$

$B \rightarrow b$

Replace all terminals with non-terminals. Any choice of capital letter.

---

e.g

$S \rightarrow ABa$

$A \rightarrow aab$

$B \rightarrow Ac$

$X \rightarrow a$ $Z \rightarrow c$

$Y \rightarrow b$

↓

Variable

Just Substitute a variable in place of terminals.

$S \rightarrow ABX$

$A \rightarrow XXY$ $B \rightarrow AZ$

S and A are not in CNF also converting them to that...

$S \to \underbrace{AB}_{K} X$ $\qquad$ $A \to \underbrace{XXY}_{L}$

$K \to AB$
$S \to KX$

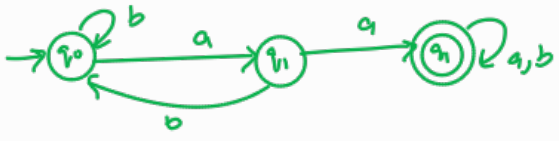
$A \to LY$
$L \to XX$

(null → unit → useless)

Ex. Transition function for DFA

$\delta^*(q, \wedge) \to q$

$\delta^*(q, ya) \to \delta(\delta^*(q, y), a)$



e.g $\delta^*(q_0, baaba)$

$\delta(\delta^*(q_0, baab), a) \to q_2$

$\delta(\delta^*(q_0, baa), b) \to q_2$

$\delta(\delta^*(q_0, ba), a) \to q_2$

$\delta(\delta^*(q_0, b), a) \to q_1$

$\delta(\delta^*(q_0, \wedge), b) \to q_0$

Ex transition for nfa